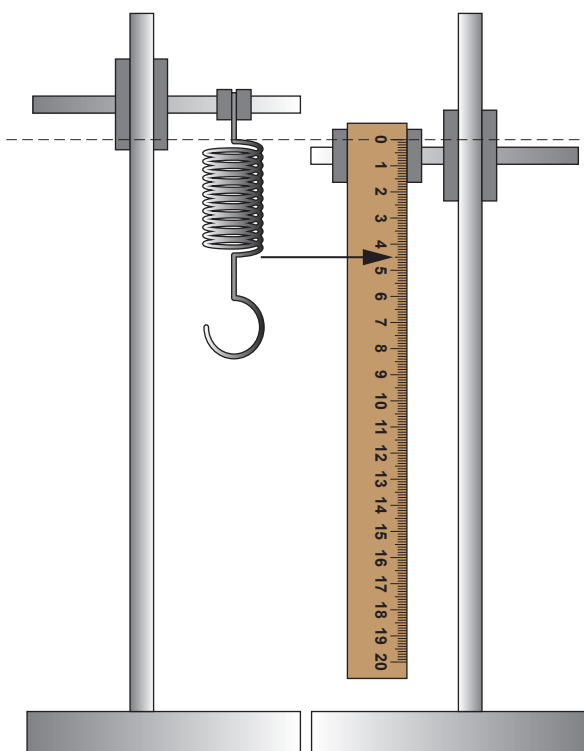
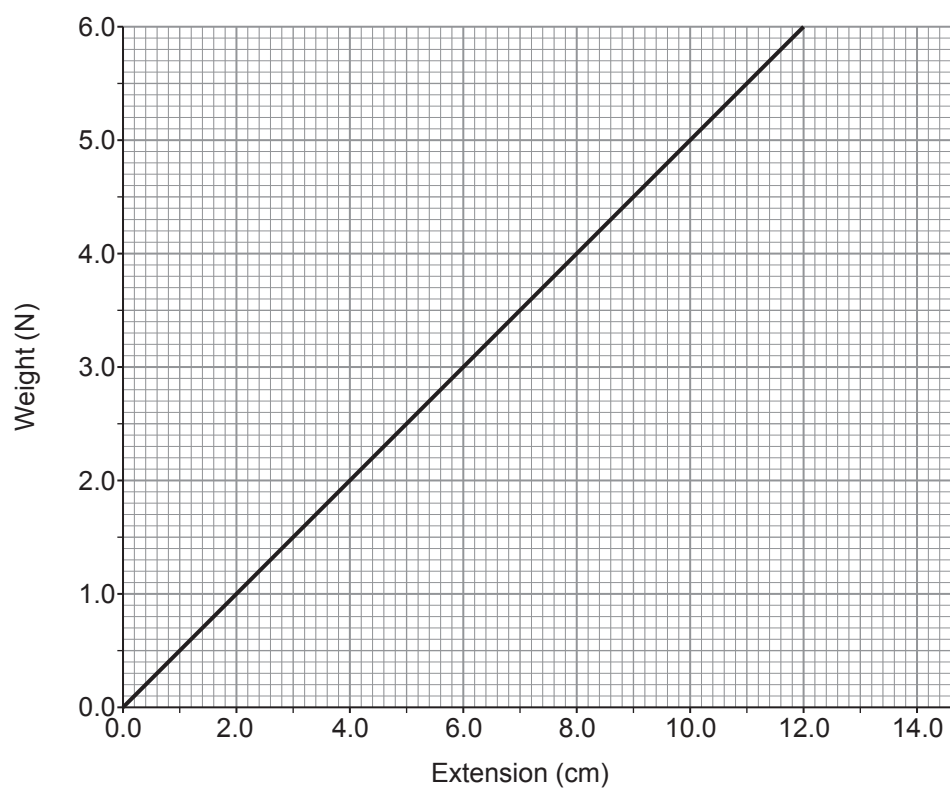
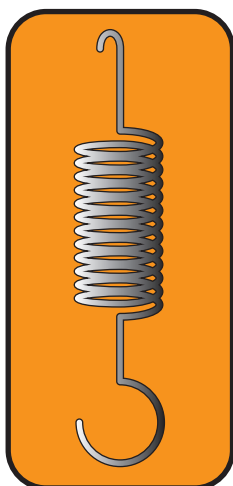
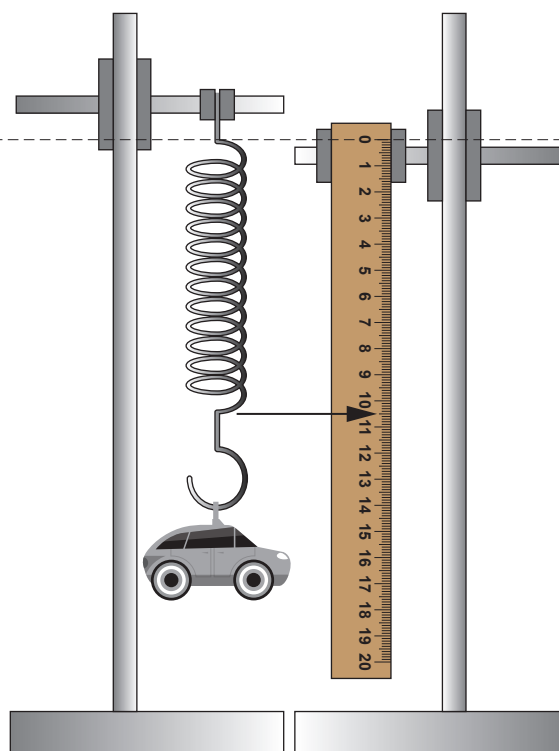


8. A school science technician bought a spring for an experiment. The data leaflet about the spring is shown below.

Data leaflet



Picture 1



Picture 2



- (a) A student used the spring to determine the mass of a toy.

Two pictures of the experiment were taken by the student.

Picture 1 shows the unstretched spring and Picture 2 shows the spring with the toy attached.

- (i) Use information from the two pictures to determine the spring's extension with the toy attached. [1]

spring's extension = cm

- (ii) Use information from the leaflet, and an equation from page 2, to calculate the mass of the toy. ($g = 10 \text{ N/kg}$) [3]

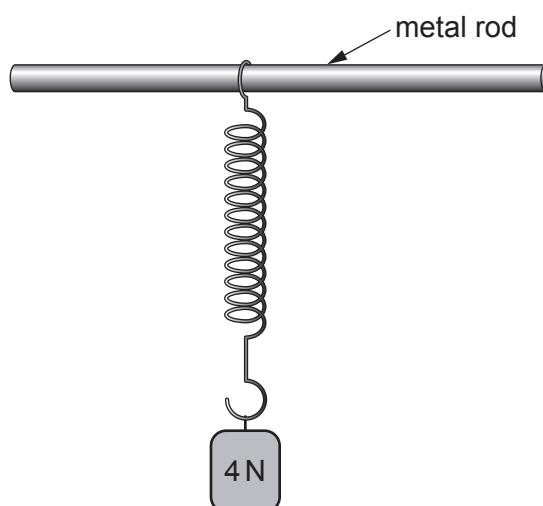
mass of toy = kg

- (b) (i) State Newton's 3rd law. [2]

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- (ii) The diagram below shows the spring suspended from a metal rod. A weight of 4 N is attached to the spring.

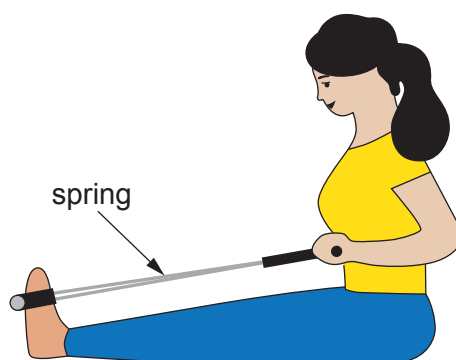


State the **size and direction** of the force that the spring applies to the weight. [2]

size = N direction =



- (c) The picture shows a woman using a piece of gym equipment containing a **different** spring.



Use the equation:

$$W = \frac{1}{2}Fx$$

to answer the following questions.

- (i) Initially the spring is unstretched.
The woman applies a force of 35 N and the spring extends 0.32 m.
Calculate the work done by the woman stretching the spring.

[2]

work done = J

- (ii) A gym instructor suggests to the woman, if she applies **double** the force to the spring she will double the amount of work done.
Assuming the spring obeys Hooke's law (applied force is directly proportional to extension) determine whether the gym instructor is correct.
Space for calculation.

[3]

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END OF PAPER

